Modelling Interventions

R_0 = Basic Reproduction #

R_c = Under control (fully susceptable population)

R_e = Effective Reproduction # (during epidemic with/without intervention)

Infections decline when $R_e < 1$

$R_c < 1$ to prevent outbreaks.

R_c depends on:

- Contact Rate
- Probability of transmitting } β
- Length time spent infectious → γ, μ etc
- Proportion of population that is susceptable → S

R_c = per person transmission rate × time infectious × proportion susceptable

$$R_c = \frac{\beta^*}{\gamma^*} S^*$$

β^*: effective transmission rate

S^*: proportion susceptable

γ^*: length time spent infectious

Factor	Control Methods
Contact Rate	Quarentine, social restrictions, "bubbles"
Transmission probability	Hygiene measures, PPE, physical distancing
Time Spent infectious	Early detection, treatment, culling, treating latent infections
proportion of pop susceptable	vaccination

Active Immunity

Natural (due to infection)

Vaccine induced

↓

Takes time to develope

Long-lasting

Passive Immunity

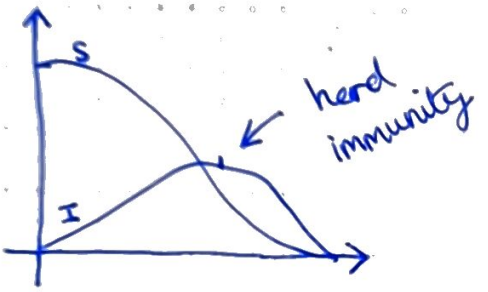
Maternal (through placenta)

Anti-body containing blood products

↓

Immediate

Short-term



$$\text{Immune} > 1 - \frac{1}{R_0}$$

ϕ = vaccine success rate

Waning Immunity

$$\text{Rate of immunity loss} = \frac{1}{\text{mean duration of immunity}}$$

convert to days